

Lily's Florist

AI Engineering Project — Final Submission

Meeting Customer Requirements

Requirements gathering, user stories, user journey, formal requirements, traceability, and annotated mockup interfaces for an AI-assisted online florist ordering experience.

Lily interview evidence • 30 user stories • customer journey • 35 requirements • traceability matrix • 7 annotated core journey mockups

1. Lily Interview Evidence

The requirements and user stories are based on the linked Lily interview/conversation and the supporting Requirements Inventory (*Lily_Interview_Notes_Step5.docx*), which separates confirmed facts, project assumptions, and open questions from the interview.

Primary interview discussion

<https://chatgpt.com/share/6ab583cf-a810-83eb-987c-9f6593724535>

Key documented business context

Lily's Florist is an established florist serving roughly 20–30 orders per day, with substantially higher demand around major occasions. Current ordering is handled through phone, in-store, and a basic online form. The project therefore focuses on dependable online ordering, inventory and delivery scheduling, secure payment, customer accounts, and an AI assistant that supports — but does not control — checkout.

AI execution boundary

The AI assistant may recommend products and add recommended products to the cart. It must clearly show what it added and allow the customer to review, change, or remove items. The customer must complete checkout and payment; the AI must never place the order.

2. User Stories

The project contains 30 user stories, exceeding the brief's minimum of 15.

US-01 — Guest / New Customer

As a guest customer, I want to browse and compare available flower arrangements, so that I can find an arrangement that suits my occasion and preferences.

US-02 — Guest / New Customer

As a guest customer, I want to customize an arrangement by choosing available size, colors, message, and extras, so that I can create an arrangement that fits my needs.

US-03 — Guest / New Customer

As a guest customer, I want to see product availability and available delivery or pickup options before checkout, so that I can choose an option that can actually be fulfilled.

US-04 — Guest / New Customer

As a guest customer, I want to complete an order without creating an account, so that I can purchase flowers conveniently.

US-05 — Customer

As a customer, I want to review and change my order before paying, so that I can correct any details before placing the order.

US-06 — Customer

As a customer, I want to pay securely using supported cards or digital wallets, so that I can complete my purchase safely.

US-07 — Customer

As a customer, I want to receive confirmation and status updates for my order, so that I know whether my order has been confirmed, prepared, and delivered or collected.

US-08 — Returning Customer

As a returning customer, I want to save recipient details, addresses, and favorite arrangements, so that I can place future orders more easily.

US-09 — Returning Customer

As a returning customer, I want to view my previous orders and reorder an arrangement, so that I can quickly repeat a successful purchase.

US-10 — Returning Customer

As a returning customer, I want to earn and redeem loyalty rewards, so that I can receive benefits from continued purchases and referrals.

US-11 — Customer — AI Support

As a customer, I want the AI assistant to answer questions about flowers, arrangements, prices, care, occasions, availability, delivery, and promotions, so that I can get help without waiting for staff.

US-12 — Customer — AI Support

As a customer, I want the AI assistant to recommend arrangements and complementary products based on my occasion, budget, colors, and preferences, so that I can make a more informed choice.

US-13 — Customer — AI Support

As a customer, I want the AI assistant to add recommended products to my cart while allowing me to review everything myself, so that I can use AI assistance without giving it control of checkout or payment.

US-14 — Customer — AI Support

As a customer, I want to communicate with the AI assistant in English or Irish, so that I can use the service in my preferred language.

US-15 — Customer — AI Support

As a customer, I want unusual, uncertain, custom, or complaint-related requests to be handed to staff, so that I receive accurate help instead of an AI-generated guess.

US-16 — Staff

As a staff member, I want to view and update order statuses, delivery or pickup information, and relevant customer details, so that I can manage orders accurately.

US-17 — Staff

As a staff member, I want to view inventory quantities and low-stock or unavailable items clearly, so that I can identify stock issues before accepting orders.

US-18 — Staff

As a staff member, I want to record inventory received, used, damaged, or unavailable quantities, so that stock information remains accurate.

US-19 — Manager

As a manager, I want to create and manage arrangements and prices without technical assistance, so that the website reflects the florist's current products and pricing.

US-20 — Manager

As a manager, I want to create, schedule, edit, and end promotions without technical assistance, so that seasonal offers remain current.

US-21 — Manager

As a manager, I want to manage seasonal content and holiday order limits without technical assistance, so that the website reflects current business conditions.

US-22 — Customer

As a customer, I want to see whether same-day delivery is available before ordering, so that I know whether my order can be delivered on the day I need it.

US-23 — Customer

As a customer, I want to subscribe to recurring flower deliveries with my preferred schedule, duration, size, and style, so that I can receive flowers regularly without placing each order manually.

US-24 — Customer

As a customer, I want to purchase and redeem digital gift cards, so that I can give or use a flexible gift option through the website.

US-25 — Customer

As a wedding or event customer, I want to submit my event date, venue, guest count, floral needs, budget, style, and inspiration photos, so that Lily can review my request and prepare a personalized quote.

US-26 — Customer

As a customer, I want to contact the florist through available human support channels and have my inquiry recorded, so that I can receive help for questions or issues that require staff.

US-27 — Staff

As a staff member, I want to manage delivery capacity and record failed-delivery outcomes, so that delivery schedules remain accurate and customers can be contacted for redelivery or pickup when needed.

US-28 — Manager

As a manager, I want to view reports on sales, order volume, busy periods, popular arrangements, inventory, deliveries, and loyalty activity, so that I can monitor business operations and make informed decisions.

US-29 — Owner / Admin

As the owner or administrator, I want to control staff permissions and access to sensitive information and settings, so that each user can access only what their role requires.

US-30 — Customer

As a customer, I want to request access to or deletion of my personal information where appropriate, so that I can exercise my privacy rights.

3. User Journey

Step 1 — Major Customer Goal

Customer wants to purchase flowers for a birthday using AI assistance.

Step 2 — Persona, Scenario, Goal

Persona: Guest / New Customer

Scenario: A customer needs to purchase flowers for a birthday and uses the website, including the AI assistant, to help choose an appropriate arrangement.

Goal: Find, customize, and purchase a suitable birthday arrangement with the desired delivery option, then receive confirmation.

Step 3 — Happy Path

Home → Browse/Search arrangements → AI recommends a suitable birthday arrangement → Customer selects product → Customize arrangement → Add to cart → Review cart → Choose delivery date/time → Enter recipient/order details → Payment → Order confirmation

Step 4 — Alternative Paths

Alternative 1 — Product unavailable: Customer selects a recommended arrangement → Product unavailable → AI suggests an available alternative → Customer selects alternative → Continue to customization/cart

Alternative 2 — Delivery slot unavailable: Customer chooses delivery → Preferred date/time unavailable → System shows available slots → Customer selects another slot → Continue to checkout

Alternative 3 — Customer changes the recommendation: AI adds recommended item to cart → Customer reviews cart → Customer changes or removes item → Continue with revised cart

Step 5 — Failure Scenarios

Failure 1 — Payment fails: Customer submits payment → Payment fails → Clear payment-error message → Customer retries or chooses another payment method → Payment succeeds → Order confirmation

Failure 2 — Order/payment status cannot be confirmed: Payment attempt occurs → Status is uncertain → System does not incorrectly confirm the order → Order/payment status is reconciled accurately → Customer receives the correct confirmation/status

Failure 3 — Delivery fails: Order is out for delivery → Delivery fails → Staff records the reason → Customer is contacted → Redelivery or pickup is arranged

AI Interaction Points

AI recommends a suitable birthday arrangement, can add recommendations to the cart, and can suggest an available alternative when a selected product is unavailable. The customer remains responsible for reviewing the cart and completing checkout and payment.

Core journey represented by the mockups: Home → Browse/Search → AI recommendation → Product Details → Cart → Checkout → Confirmation. Delivery/pickup selection is represented within Product Details and Checkout rather than as a separate screen.

4. Formal Requirements

The requirements document contains 27 functional requirements and 8 non-functional requirements (35 total).

Functional Requirements

FR-01 — Product Catalog

The system shall allow customers to browse products and view photos, prices, sizes, flower types, colors, and occasions.

FR-02 — Product Customization

The system shall allow customers to customize eligible arrangements by selecting available size, colors, message, and extras before adding them to the cart.

FR-03 — Product Availability

The system shall display current product availability and prevent customers from ordering products that are unavailable.

FR-04 — Delivery and Pickup Scheduling

The system shall allow customers to select available delivery or pickup dates and time slots, including same-day delivery when available and requested by the 1 PM cutoff.

FR-05 — Checkout

The system shall allow customers to complete an order as a guest or through an account.

FR-06 — Order Review

The system shall allow customers to review and change order details before payment.

FR-07 — Payment Methods

The system shall support card payments, Apple Pay, and Google Pay.

FR-08 — Payment Failure Handling

The system shall display a clear message when payment fails and allow the customer to retry or use another supported payment method.

FR-09 — Confirmation and Order Status

The system shall provide confirmation after successful payment and allow customers to view order status and receive status updates.

FR-10 — Customer Account and Loyalty

The system shall allow returning customers to save recipient details, addresses, and favorites; view previous orders and reorder; and earn, view, and redeem loyalty rewards.

FR-11 — AI Information Support

The system shall allow the AI assistant to answer questions about flowers, arrangements, prices, care, occasions, availability, delivery, and promotions.

FR-12 — AI Recommendations

The system shall allow the AI assistant to recommend arrangements and complementary products based on the customer's occasion, budget, colors, and available preferences.

FR-13 — AI Cart Addition

The system shall allow the AI assistant to add recommended products to the customer's cart and clearly identify the items it added.

FR-14 — Customer Control of AI Cart Items

The system shall allow customers to review, change, or remove AI-added items before checkout, and shall prevent the AI assistant from placing the order or completing payment.

FR-15 — AI Language Support

The system shall allow customers to communicate with the AI assistant in English or Irish while using the same product, pricing, availability, and delivery information.

FR-16 — AI Escalation

The system shall allow the AI assistant to escalate uncertain, unusual, custom, or complaint-related requests to staff.

FR-17 — Staff Order Management

The system shall allow authorized staff to view and update order statuses, delivery or pickup information, and relevant customer details.

FR-18 — Inventory Management

The system shall allow authorized staff to view inventory quantities and low-stock or unavailable items and record quantities received, used, damaged, or unavailable.

FR-19 — Product and Price Management

The system shall allow authorized managers to create and manage arrangements and prices without technical assistance.

FR-20 — Promotion Management

The system shall allow authorized managers to create, schedule, edit, and end promotions without technical assistance.

FR-21 — Seasonal Content and Holiday Limits

The system shall allow authorized managers to manage seasonal content and holiday order limits without technical assistance.

FR-22 — Subscriptions

The system shall allow customers to subscribe to recurring flower deliveries with a preferred schedule, duration, size, and style.

FR-23 — Gift Cards

The system shall allow customers to purchase and redeem digital gift cards.

FR-24 — Event Inquiries

The system shall allow wedding or event customers to submit event date, venue, guest count, floral needs, budget, style, and inspiration photos for Lily's review.

FR-25 — Human Support

The system shall allow customers to submit human-support inquiries through available support channels and allow staff to record and track those inquiries.

FR-26 — Delivery Management

The system shall allow authorized staff to manage delivery capacity and record failed-delivery outcomes so customers can be contacted for redelivery or pickup.

FR-27 — Management Reporting

The system shall allow authorized managers to view reports covering sales, order volume, busy periods, popular arrangements, inventory, deliveries, and loyalty activity.

Non-Functional Requirements

NFR-01 — Page Performance

The system shall load normal customer-facing pages in approximately 2 seconds under expected operating conditions (exact acceptance threshold: TBD).

NFR-02 — Checkout Performance

The system shall complete checkout actions in approximately 3 seconds under expected operating conditions (exact acceptance threshold: TBD).

NFR-03 — Peak-Period Reliability

The system shall support peak periods of 100+ orders per day without timeouts or failed orders (detailed load-test threshold: TBD).

NFR-04 — Payment and Order Consistency

The system shall not confirm an order unless payment succeeds and shall not charge a customer without a corresponding confirmed transaction.

NFR-05 — Security and Role-Based Access

The system shall provide secure authentication, temporary lockout after several failed login attempts (exact threshold: TBD), secure password reset through the registered email address, and role-limited access to staff, manager, and owner/admin information and settings.

NFR-06 — Privacy and Data Protection

The system shall protect customer personal data in accordance with applicable Irish/EU privacy requirements and support appropriate requests for access to or deletion of personal information.

NFR-07 — Accessibility and Responsive Use

The system shall provide readable content, appropriate contrast, image descriptions, simple navigation, assistive-technology-friendly forms, and a usable experience across mobile, tablet, and desktop devices.

NFR-08 — Maintainability, Hosting, and Backups

The system shall use a dependable hosted service that supports reliable operation during high-demand periods and regular backups, and shall allow Lily and authorized managers to manage relevant business content without technical assistance. Exact backup frequency, retention, and recovery targets remain open questions (TBD).

Composite requirements note: FR-10 and FR-25 intentionally combine related capabilities retained together to preserve the existing ID sequence; each listed capability should be tested separately during QA.

5. Documented Open Questions

The source materials identify the following items as unresolved. No values have been invented for them.

1. Delivery area / ZIP or postcode coverage.
2. Exact delivery-fee calculation rules.
3. Exact loyalty earning and redemption rules.
4. Gift-card expiry and value limits.
5. Accessibility compliance level, such as a specific WCAG target.
6. Backup frequency, retention, and recovery target.
7. Specific payment and email service providers.
8. Exact holiday order-cap limits.

These open questions remain intentionally unresolved in the formal requirements.

6. Traceability Matrix

The matrix preserves the original user-story and requirement sequence. **Note:** entries reading "Supporting interface (...)" identify a required capability only — they are not part of the seven submitted core mockups and are not visual evidence of a built screen.

US ID	Requirement ID	Type	Visual Evidence / Interface Scope
US-01	FR-01	Functional	Home; Product Search; Product Details
US-02	FR-02	Functional	Product Details
US-03	FR-03; FR-04	Functional	Product Details; Checkout
US-04	FR-05	Functional	Checkout
US-05	FR-06	Functional	Cart; Checkout
US-06	FR-07; FR-08	Functional	Checkout
US-07	FR-09	Functional	Confirmation
US-08	FR-10	Functional	Supporting (Account) — not a core mockup
US-09	FR-10	Functional	Product Details; Supporting (Account)
US-10	FR-10	Functional	Supporting (Account); Checkout
US-11	FR-11	Functional	AI Assistant
US-12	FR-12	Functional	AI Assistant; Product Details
US-13	FR-13; FR-14	Functional	AI Assistant; Cart
US-14	FR-15	Functional	AI Assistant
US-15	FR-16	Functional	AI Assistant
US-16	FR-17	Functional	Supporting (Staff/Admin)
US-17	FR-18	Functional	Supporting (Staff/Admin)
US-18	FR-18	Functional	Supporting (Staff/Admin)
US-19	FR-19	Functional	Supporting (Staff/Admin)
US-20	FR-20	Functional	Supporting (Staff/Admin)
US-21	FR-21	Functional	Supporting (Staff/Admin)
US-22	FR-04	Functional	Product Details; Checkout
US-23	FR-22	Functional	Product Details; Checkout; Supporting (Account)
US-24	FR-23	Functional	Checkout; Supporting (Account)
US-25	FR-24	Functional	Home; Supporting (Contact)
US-26	FR-25	Functional	Supporting (Contact)
US-27	FR-26	Functional	Supporting (Staff/Admin)
US-28	FR-27	Functional	Supporting (Staff/Admin)
US-29	NFR-05	Non-Functional	Supporting (Staff/Admin)
US-30	NFR-06	Non-Functional	Supporting (Account); Supporting (Contact)

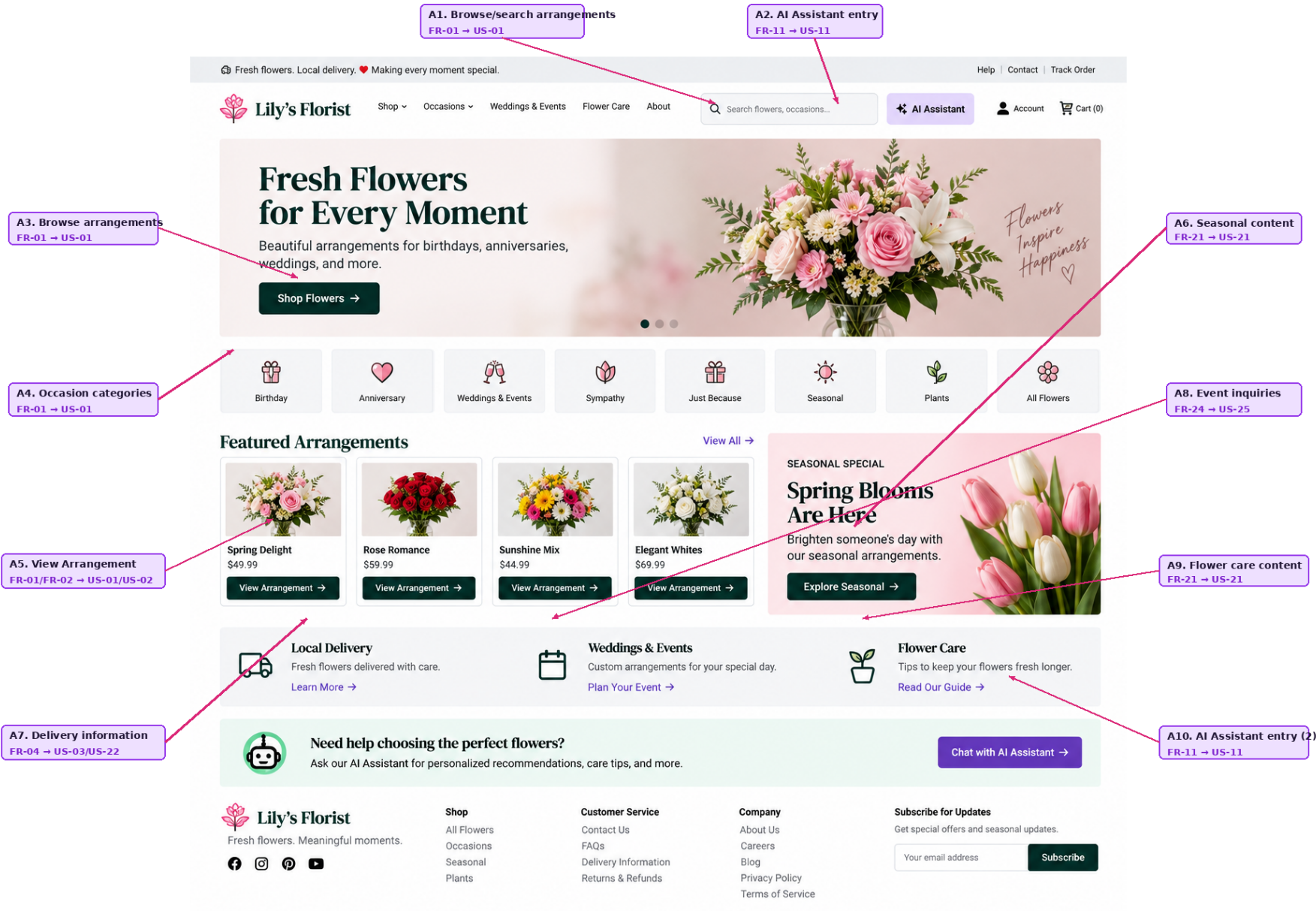
Core mockup scope: Home, Product Search, AI Assistant, Product Details, Cart, Checkout, and Confirmation. Supporting Account, Contact, and Staff/Admin interfaces are documented for traceability but are not part of the seven core journey mockups.

7. Annotated Mockup Interfaces

The following seven annotated interfaces represent the core customer journey. Each screen includes requirement/story markers, a short design-rationale note, and traceability back to the formal requirements.

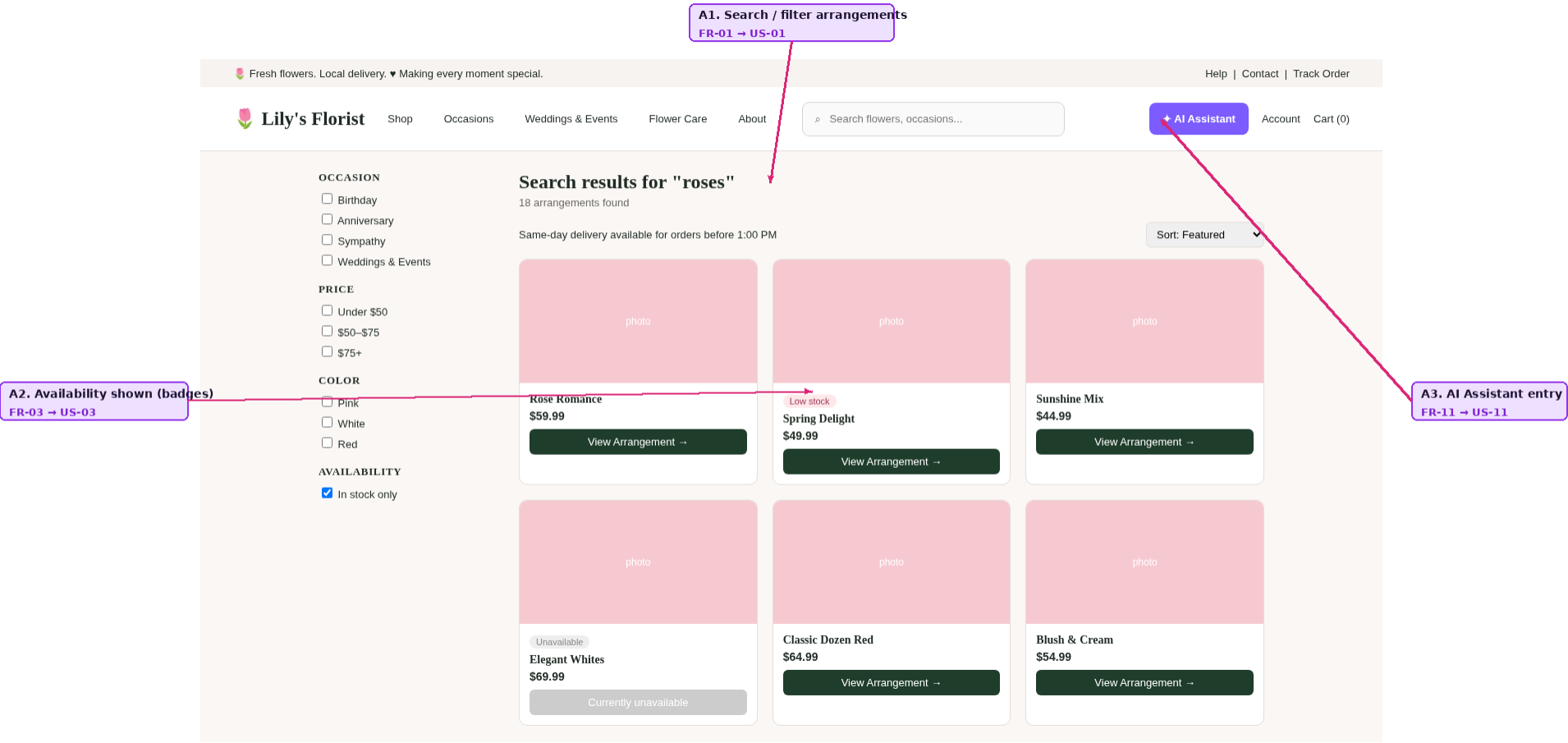
01 — Home

Design rationale: 'View Arrangement' (not 'Add to Cart') is used on Home because customization (FR-02) must happen on Product Details before an item reaches the cart, per the approved user journey.



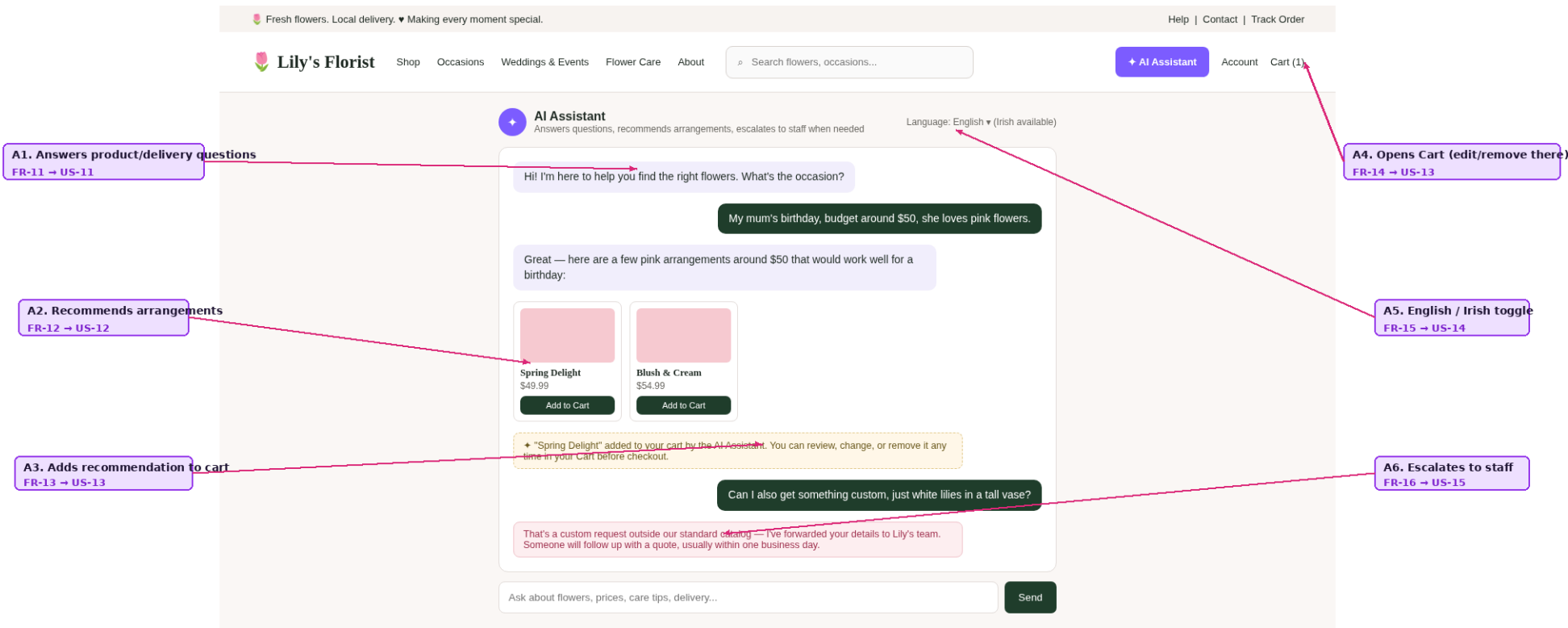
02 — Product Search

Design rationale: Availability badges ('Low stock' / 'Unavailable') are shown inline on results, not only on Product Details, so customers can filter out unfulfillable items before clicking through (FR-03).



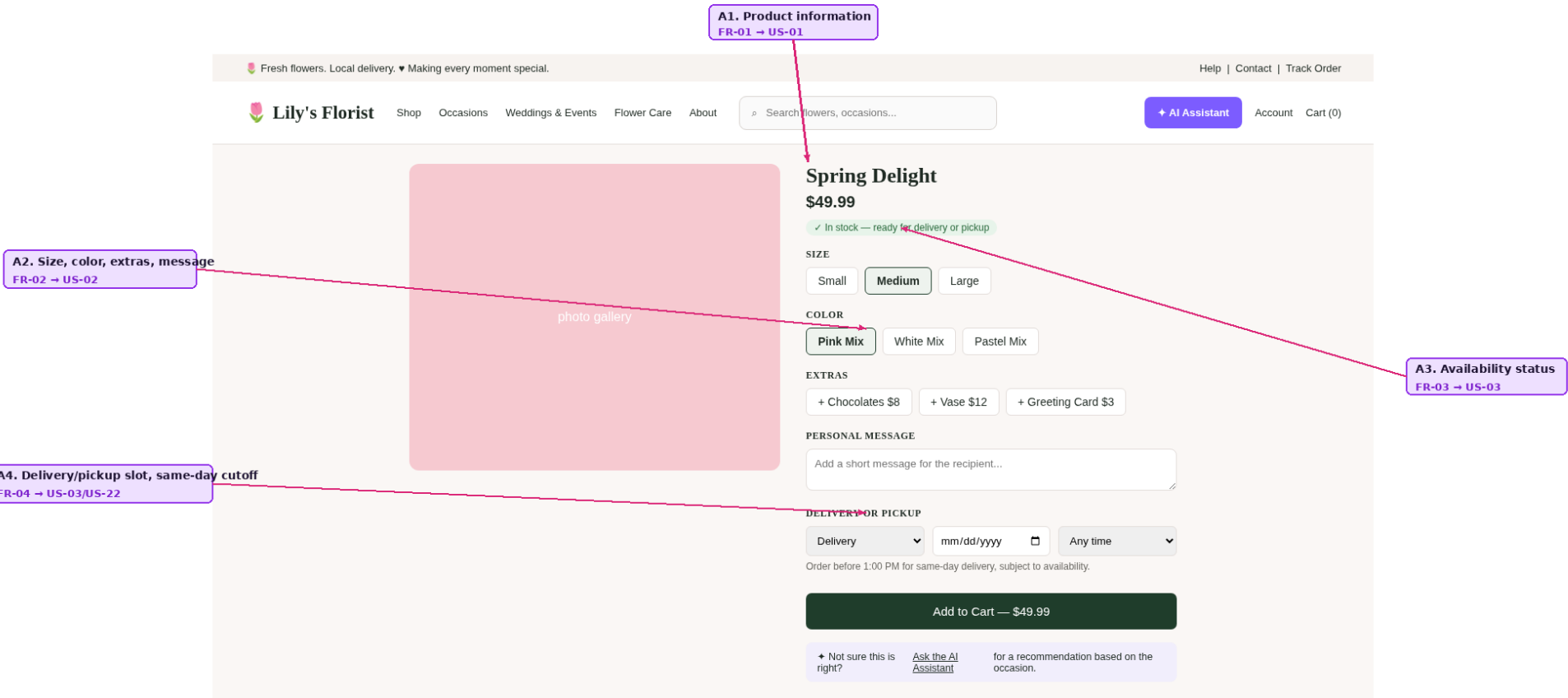
03 — AI Assistant

Design rationale: The yellow banner (A3) shows what the AI added; full review/edit/remove controls live on the Cart screen (A1/A2), not here (FR-14).



04 — Product Details

Design rationale: Customization (A2) sits above delivery scheduling (A4) so the customer configures the product before committing to a delivery window, matching the FR-02 → FR-04 order in the happy path.



05 — Cart

Design rationale: The AI-added badge (A1) and Edit/Remove controls (A2) are on the same line as every other cart item, so AI-sourced items are visible but not treated as locked or special — the customer has full control (FR-14).

Fresh flowers. Local delivery. ♥ Making every moment special.

Help | Contact | Track Order

Lily's Florist

Shop

Occasions

Weddings & Events

Flower Care

About

Search flowers, occasions...

+ AI Assistant

Account

Cart (2)

Your Cart

+ Added by AI Assistant

Spring Delight — Medium, Pink Mix

Message: "Happy Birthday Mum!" · Delivery: Sat, 27 Sep, 12–3PM

Edit Remove

\$49.99

Digital Gift Card

Value: \$25 · Delivered by email

Edit Remove

\$25.00

Order Summary

Subtotal\$74.99

Delivery fee\$6.00

Promo code

Apply

Total\$80.99

Proceed to Checkout →

A1. AI-added item identified
FR-14 → US-13

A2. Edit / remove any item
FR-14 → US-13

A3. Review before checkout
FR-06 → US-05

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06 — Checkout

Design rationale: 'Place Order' stays disabled-feeling until payment succeeds, and the note under the button restates NFR-04: no confirmation and no charge happen until payment is actually confirmed.

Fresh flowers. Local delivery. Making every moment special.

Help | Contact | Track Order

ShopOccasionsWeddings & EventsFlower CareAbout

Search flowers, occasions...

+ AI AssistantAccountCart (0)

Checkout

1. Order Type

Guest Checkout

Sign in / Create Account

Full name

Email

2. Delivery Details

Delivery

Store Pickup

Recipient name

Phone

Delivery address

mm/dd/yyyy

12-3PM

3. Payment

Card

Apple Pay

Google Pay

Card number

Expiry / CVV

Payment could not be processed. Please try again or choose another method.

Order Summary

Spring Delight (Medium)

\$49.99

Digital Gift Card

\$25.00

Delivery fee

\$6.00

Total

\$80.99

Place Order

Your order is not confirmed and no payment is taken until this step succeeds.

A1. Guest checkout

FR-05 → US-04

A2. Delivery date/time selection

FR-04 → US-03/US-22

A3. Card/Apple/Google Pay

FR-07 → US-06

A4. Payment failure + retry

FR-08 → US-06

07 — Confirmation

Design rationale: This screen only renders after payment succeeds (see Checkout A4/NFR-04), so its mere existence in the flow is itself evidence that the order/payment consistency rule was enforced.

✓

Order Confirmed

Order #LF-20482 · Confirmation sent to your email

Spring Delight (Medium, Pink Mix)	\$49.99
Digital Gift Card	\$25.00
Delivery — Sat, 27 Sep, 12–3PM	\$6.00
Total Paid	\$80.99

Confirmed

Preparing

Out for Delivery

Delivered

Track Order

Create Account to Save Details

Want flowers for another occasion? Ask the AI Assistant to remind you before your next event.

Fresh flowers. Local delivery. Making every moment special.

Help | Contact | Track Order

Lily's Florist

Shop Occasions Weddings & Events Flower Care About

Search flowers, occasions...

AI Assistant Account Cart (0)

A1. Order confirmation & summary

FR-09 → US-07

A2. Order status tracker

FR-09 → US-07

A3. Payment/order accuracy

NFR-04 → US-06/US-07

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8. Final QA / Submission Scope

The project package contains the required evidence for the brief: the Lily interview link, 30 user stories, the user journey, 35 formal requirements, traceability, and annotated mockup interfaces.

The seven core mockups follow the documented journey from Home through Confirmation, each annotated with both requirement/story traceability and a short design-rationale note. The traceability matrix does not claim that Account, Contact, or Staff/Admin are among those seven core screens.

The final deliverable is this single PDF. A separate .fig working file is not included because the official submission instructions require the mockup interfaces in the PDF and do not explicitly require a .fig file.

Final file: Lily_Florist_AI_Project_FINAL.pdf